Your grade: 96%

Your latest: **96%** • Your highest: **96%** • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed a good understanding of data splitting, overfitting, and the forward pass in neural networks. However, there is room for improvement in evaluating neural network performance. Before retrying this assignment, we recommend focusing on this key area:

- **Neural Network from Scratch Model Evaluation**: In Question 3, your understanding of the metrics used to evaluate a neural network's performance was incomplete. Try reviewing this item to refresh your memory.

Good job.

Retry assignment

1. What is the primary purpose of splitting data into training, validation, and test sets? — 1 / 1 point

- ( ) To simplify the training process.
- ( ) To reduce the computational cost of training the model.
- ( ) To increase the model's accuracy on the training data.
- (•) To evaluate the model's performance on unseen data.

Correct
Correct! Splitting the data helps in assessing how well the model will perform on new, unseen data.

2. Which of the following are signs of a model that is overfitting? — 1 / 1 point

- [ ] The model performs equally well on both training and validation data.
- [x] The model has high variance.

Correct
Right! High variance indicates that the model is sensitive to fluctuations in the training data, a common sign of overfitting.

- [x] The model performs well on training data but poorly on validation data.

Correct
Correct! Overfitting occurs when the model captures noise in the training data, leading to poor performance on validation data.

- [ ] The model's performance improves as more noise is added to the training data.

3. Which of the following metrics can be used to evaluate a neural network's performance? — 0.8 / 1 point

- [ ] Loss function value
- [x] Confusion matrix

Correct
Good job! A confusion matrix provides detailed insights into the performance of a classification model.

- [ ] Training time
- [x] Accuracy

Correct
Correct! Accuracy is one of the primary metrics used to evaluate a neural network's performance.

- [ ] Learning rate

You didn't select all the correct answers

4. How is the derivative of the Sigmoid function significant in a neural network's forward pass? — 1 / 1 point

- ( ) It helps in setting the learning rate
- ( ) It helps in normalizing the input data
- (•) It is used during backpropagation to calculate gradients
- ( ) It is used to compute the activation of neurons

Correct
Well done! The derivative of the Sigmoid function is crucial during backpropagation as it helps in calculating the gradients.

5. What is the primary purpose of the forward pass in a neural network? — 1 / 1 point

- ( ) To initialize the network parameters
- (•) To calculate the output of the network
- ( ) To update the weights and biases of the network
- ( ) To evaluate the network's accuracy

Correct
Correct! The primary purpose of the forward pass is to compute the network's output based on the input data and the current weights and biases.